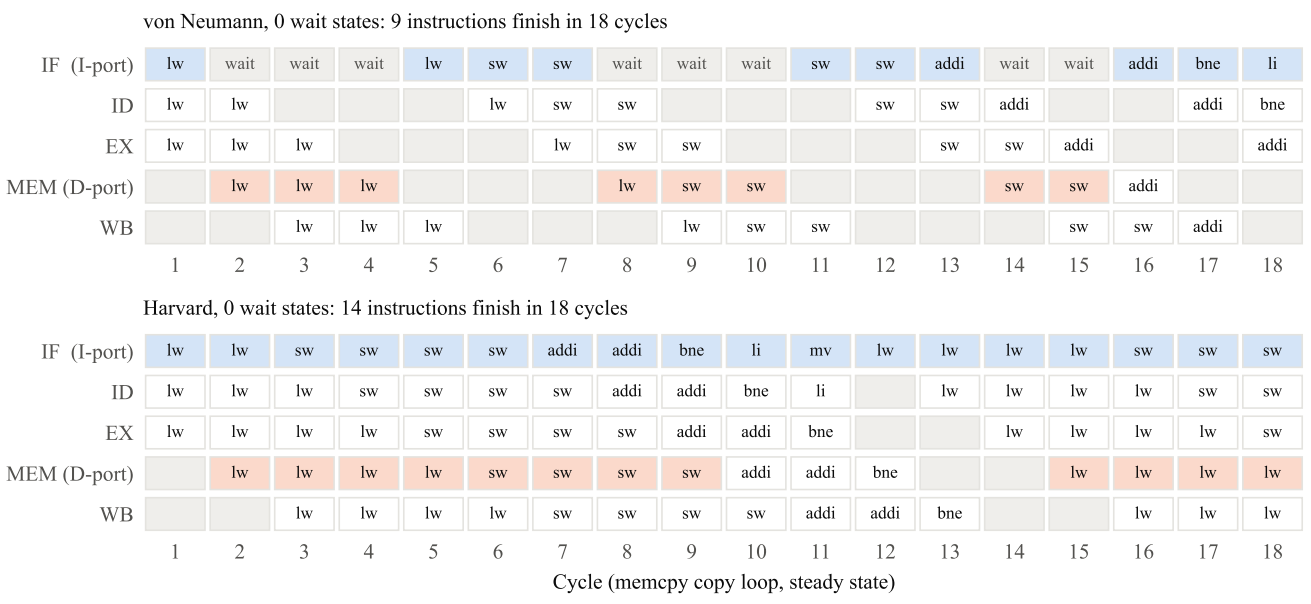
von Neumann, 0 wait states: 9 instructions finish in 18 cycles
IF (I-port) lw wait wait wait lw sw sw wait wait wait sw sw addi wait wait addi bne li
ID lw lw lw sw sw addi addi bne li lw lw addi addi bne
EX lw lw lw lw sw sw addi addi bne lw lw addi addi
MEM (D-port) lw lw lw lw sw sw sw addi addi bne lw lw lw lw sw
WB lw lw lw lw sw sw mv lw lw lw lw sw sw sw
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Harvard, 0 wait states: 14 instructions finish in 18 cycles
IF (I-port) lw lw sw sw sw sw addi addi bne li mv lw lw lw lw sw sw sw
ID lw lw lw sw sw sw sw addi addi bne li lw lw lw lw sw sw
EX lw lw lw lw sw sw sw sw addi addi bne lw lw lw lw sw
MEM (D-port) lw lw lw lw lw sw sw sw sw addi addi bne lw lw lw lw
WB lw lw lw lw lw lw sw sw sw sw addi addi bne lw lw lw
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Cycle (memcpy copy loop, steady state)
instruction fetched
data access
bubble / fetch waiting